

Clustering for Background Removal

A comparative study of K-Means, DBSCAN, and MeanShift algorithms applied to image segmentation.

Application Overview

Objective

To automatically separate the foreground subject from the background in complex images without labeled data.

Methodology

We utilize **Unsupervised Learning** (Clustering) to group pixels based on similarity. Pixels with similar color and spatial features are assigned to the same cluster.



Dataset & Features

Input Image: "tucan.jpg" - A high-resolution RGB image featuring a colorful bird against a complex natural background.

Feature Engineering

To improve clustering, we convert pixels into a 5-dimensional feature vector:

- **Color (L, a, b):** Perceptually uniform color space.
- **Position (x, y):** Spatial coordinates to enforce connectivity.



Preprocessing Pipeline



Color Conversion

Converted images from BGR (OpenCV default) to RGB for display and LAB for clustering to mimic human color perception.



Downsampling

Reduced computational load by training on a subset (e.g., 25,000 pixels) or resizing large images before clustering.



Standardization

Used StandardScaler to normalize features. This is critical for distance-based algorithms like DBSCAN.

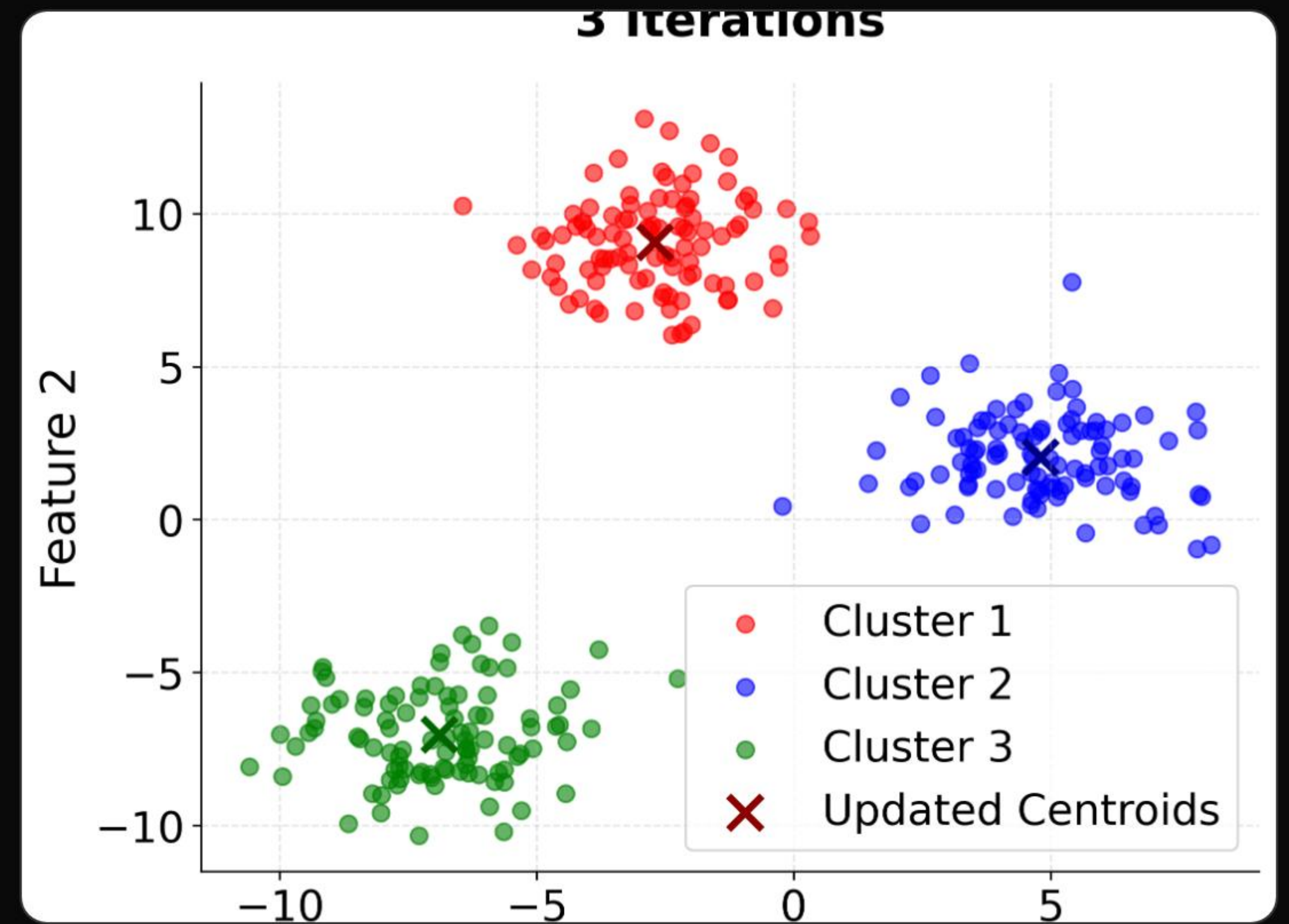
Method 1: K-Means (Manual Implementation)

Algorithm Logic

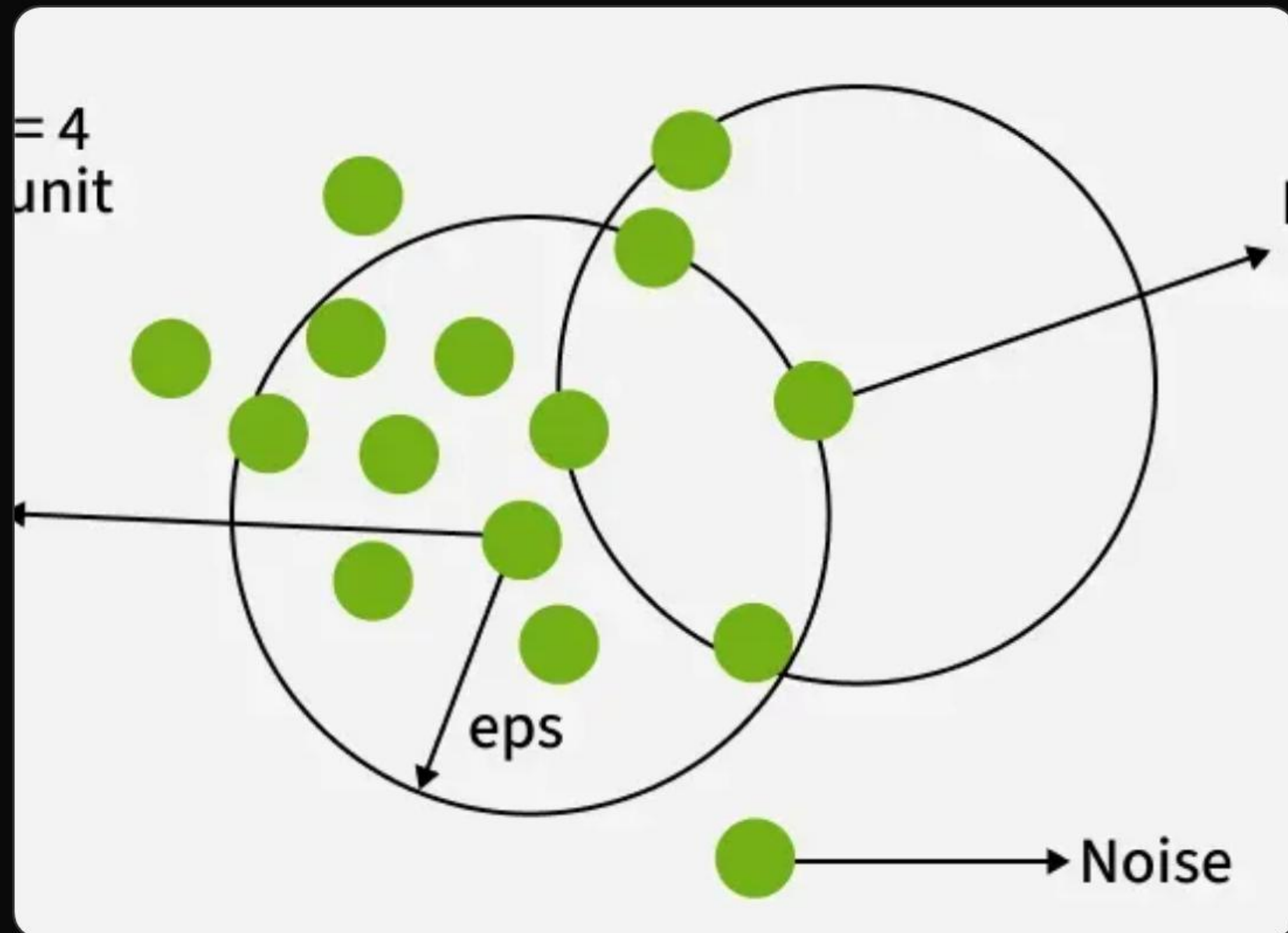
Iteratively assigns pixels to the nearest of k centroids and recalculates centroids based on the mean of assigned points.

$$J = \sum_{i=1}^k \sum_{x \in S_i} \|x - \mu_i\|^2$$

Configuration: K=3, Manual initialization, Max Iters=30.



Method 2: DBSCAN



Density-Based Clustering

Groups points that are closely packed together (points with many nearby neighbors). It finds core samples of high density and expands clusters from them.

Key Features

- ✓ **Noise Handling:** Classifies outliers as -1, effectively removing scattered background noise.
- ✓ **No fixed K:** Automatically finds the number of clusters based on `eps` (0.6) and `min_samples` (12).
- ✓ **Spatial Weighting:** Used weighted coordinates to enforce spatial coherence.

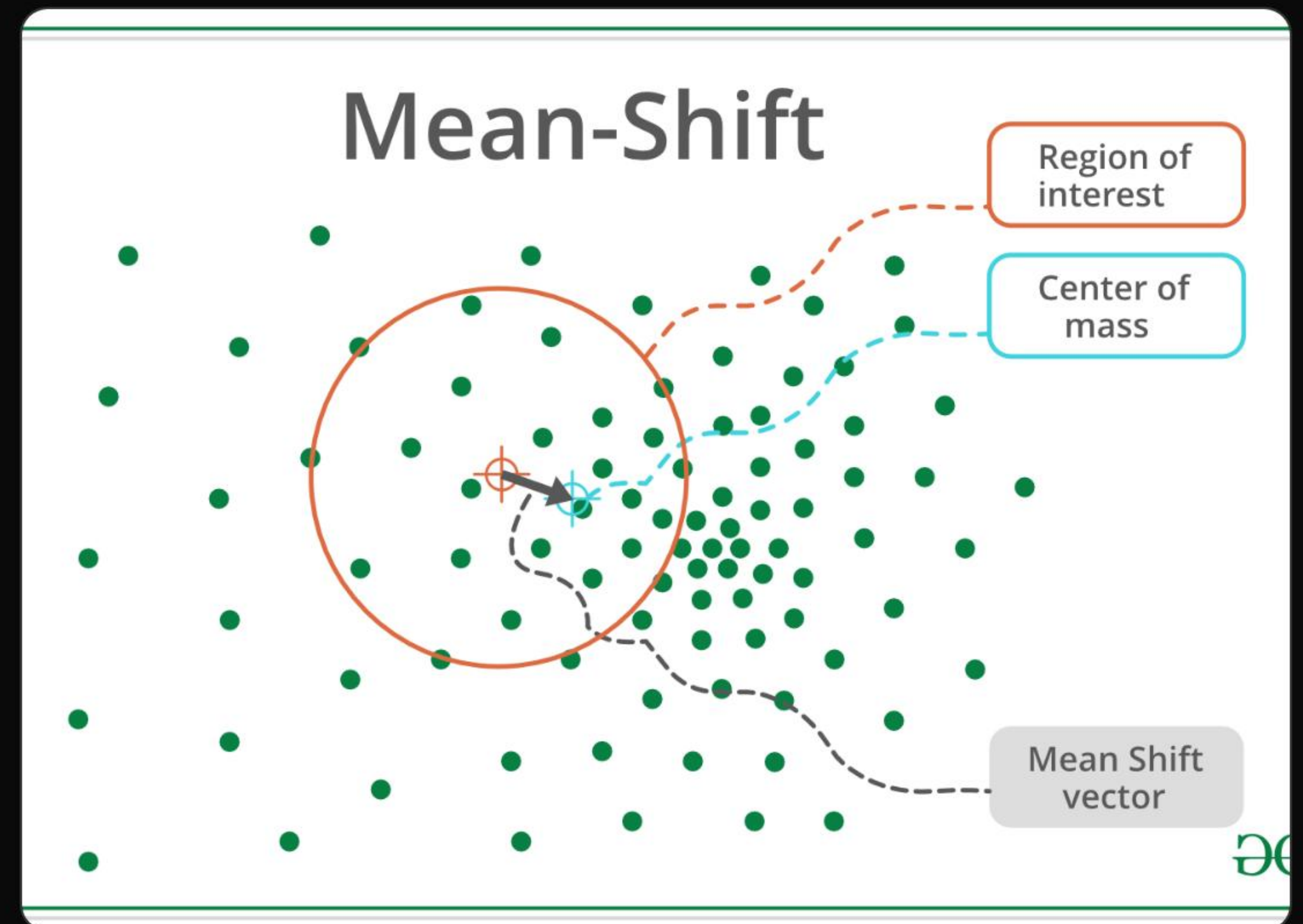
Method 3: MeanShift

Mode Seeking Algorithm

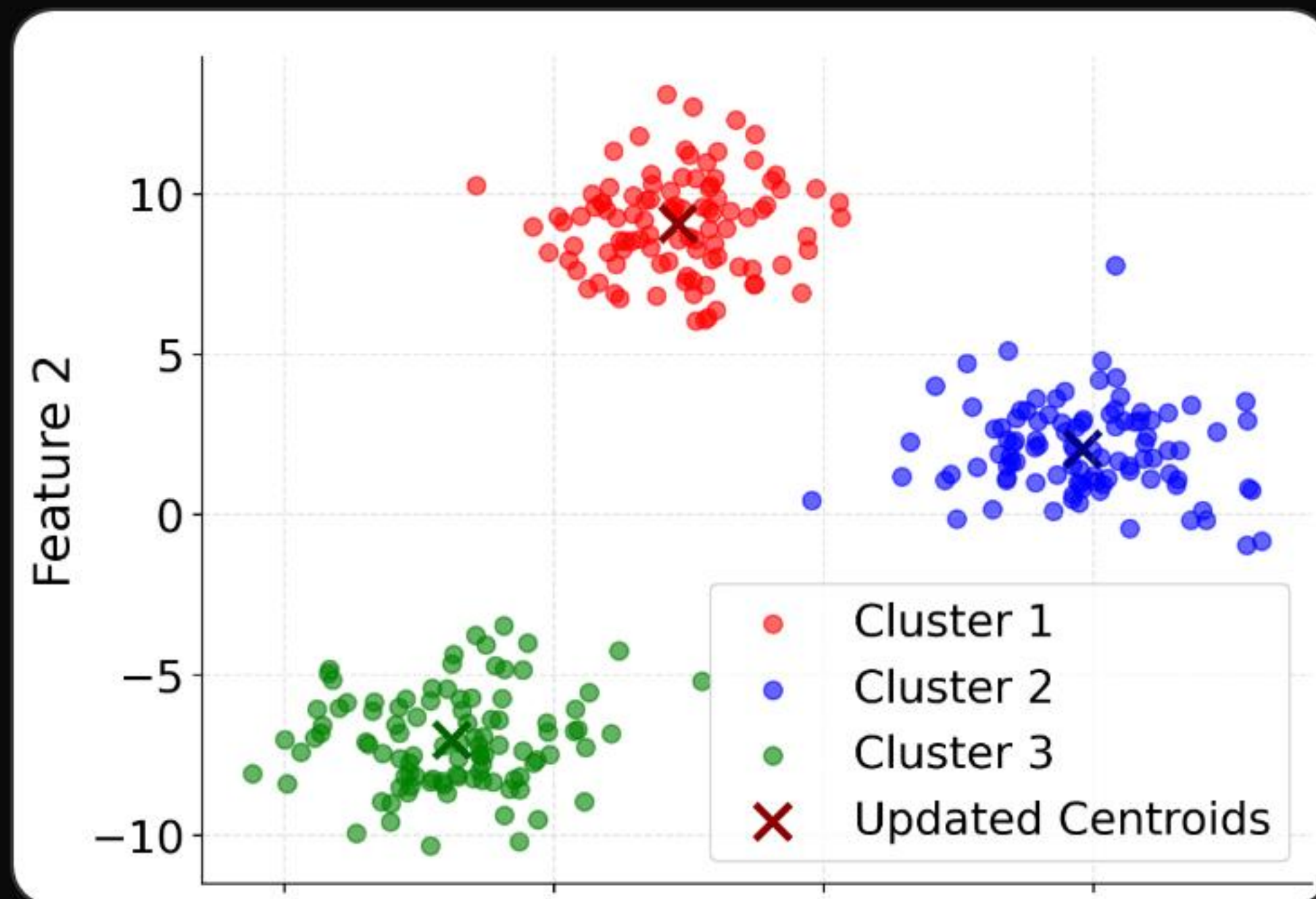
A non-parametric feature-space analysis technique for locating the maxima of a density function. It iteratively shifts data points towards the mode (highest density).

Implementation Details

- ✓ **Bandwidth Estimation:** Critical step using `estimate_bandwidth` with a quantile of 0.25.
- ✓ **Bin Seeding:** Used to speed up the convergence.
- ✓ **Result:** Produces highly smooth, "painted" look but is computationally expensive ($O(N^2)$).

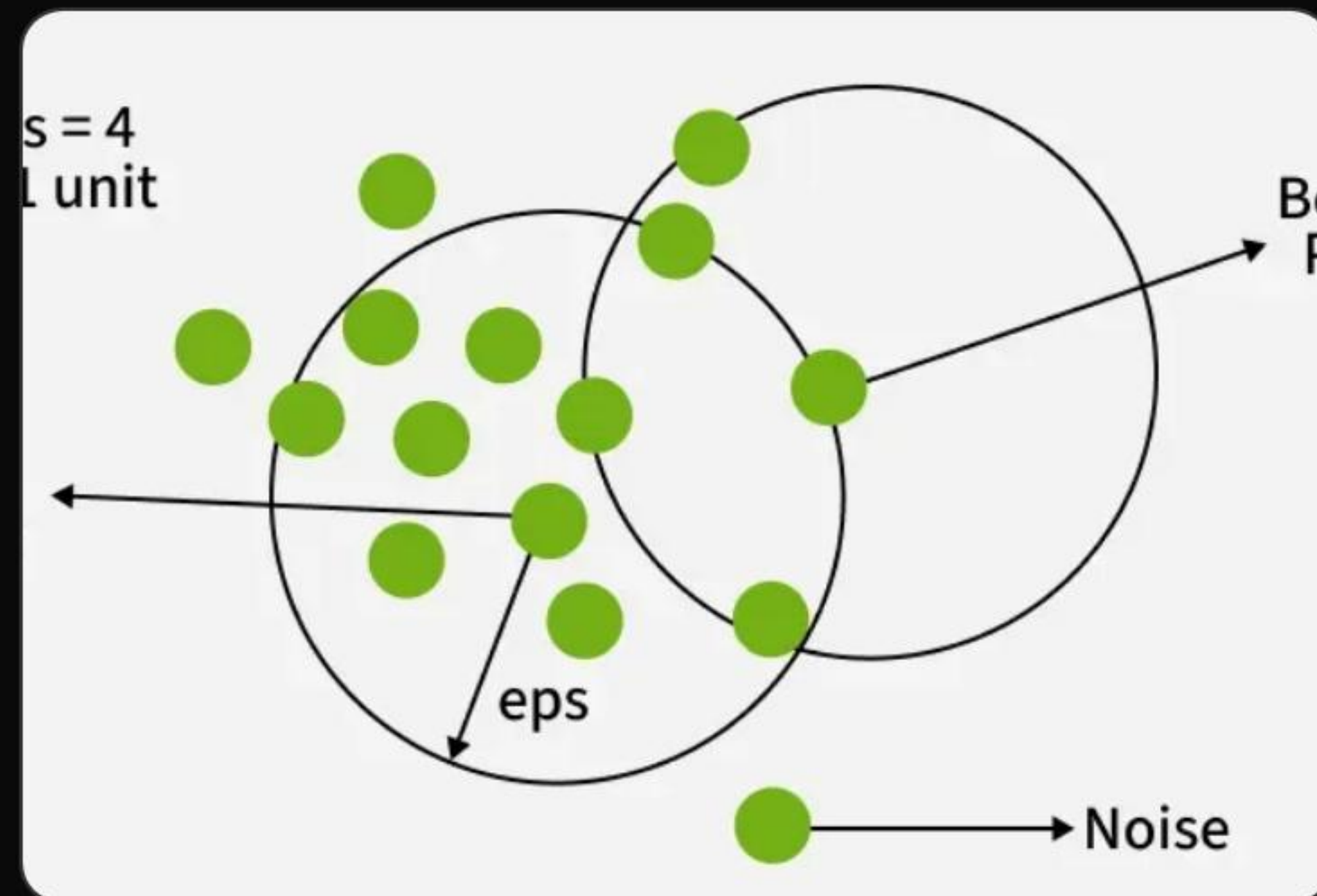


Visual Results



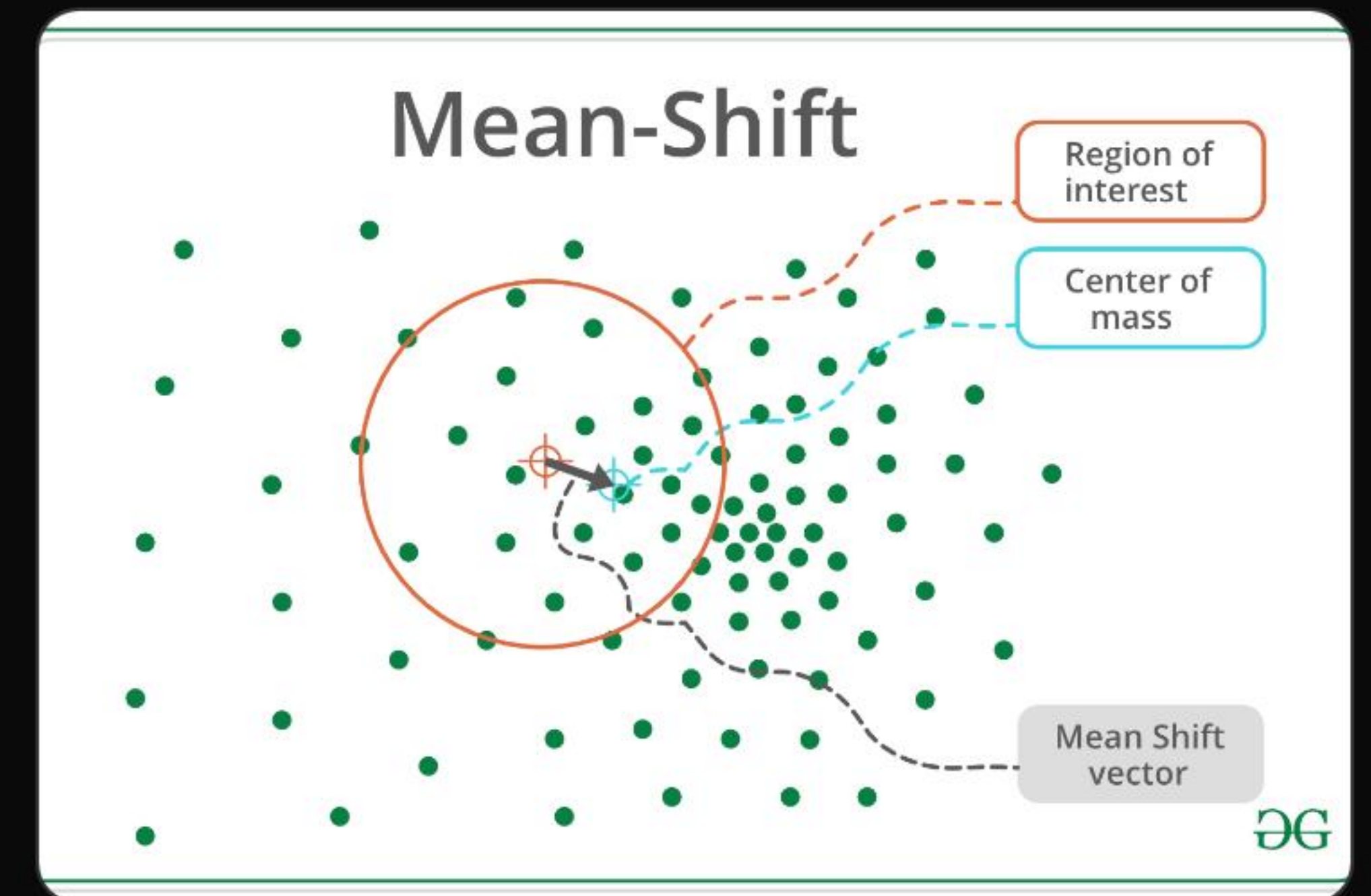
K-Means

Rigid boundaries, successful color separation.



DBSCAN

Excellent noise removal, organic shapes.



MeanShift

Smoothest segmentation, best coherence.

Quantitative Comparison

Performance metrics gathered from 25,000 sample pixels on "tucan.jpg".

Algorithm	Time (sec)	Silhouette Score	Clusters Found
K-Means	0.42s	0.65	3 (Fixed)
DBSCAN	2.10s	0.58	5 + Noise
MeanShift	14.85s	0.72	4 (Auto)

* Silhouette score indicates how similar an object is to its own cluster (cohesion) compared to other clusters (separation).

Conclusion

Trade-offs

- ✓ **Speed vs. Quality:** K-Means is the fastest but produces the roughest masks. MeanShift offers the highest quality segmentation but is too slow for real-time video.
- ✓ **Parameter Sensitivity:** DBSCAN is powerful for noise removal but requires fine-tuning of eps and feature scaling.

Final Recommendation

For static image editing where quality is paramount, **MeanShift** is the superior choice. for real-time applications or simple background subtraction, **K-Means** or Gaussian Mixture Models would be preferred.

Image Sources



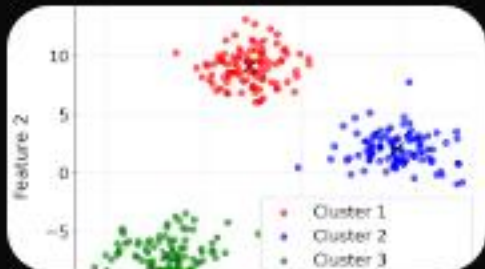
<https://inside-machinelearning.com/wp-content/uploads/2023/08/list-segmente-1024x576.png>

Source: inside-machinelearning.com



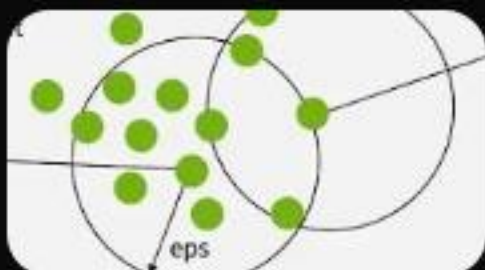
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Source: www.ukposters.co.uk



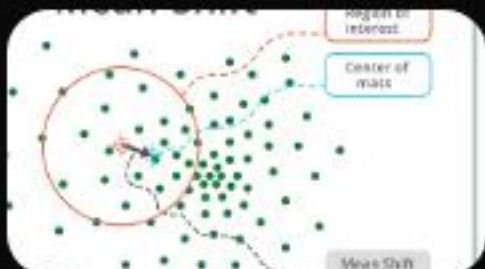
https://cnassets.uk/notebooks/kmeans_files/kmeans-algorithm-step-3-third-iteration.png

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